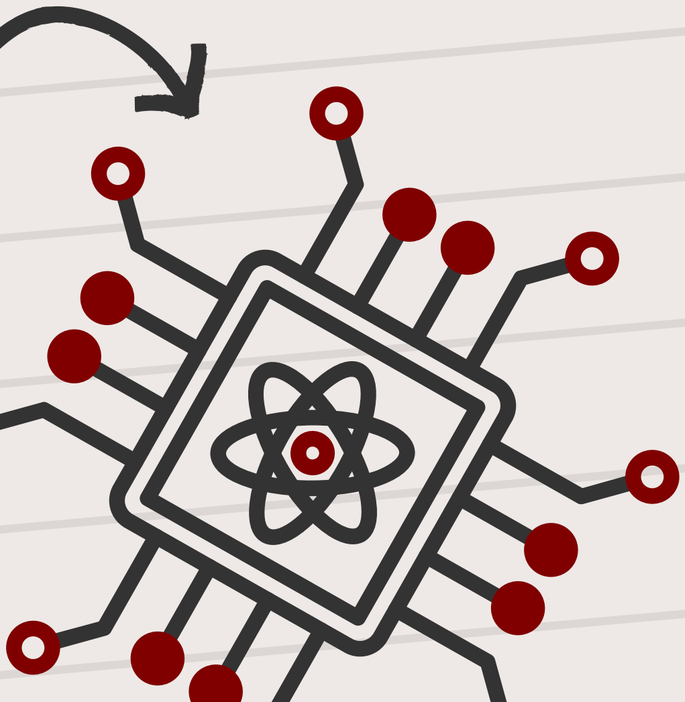
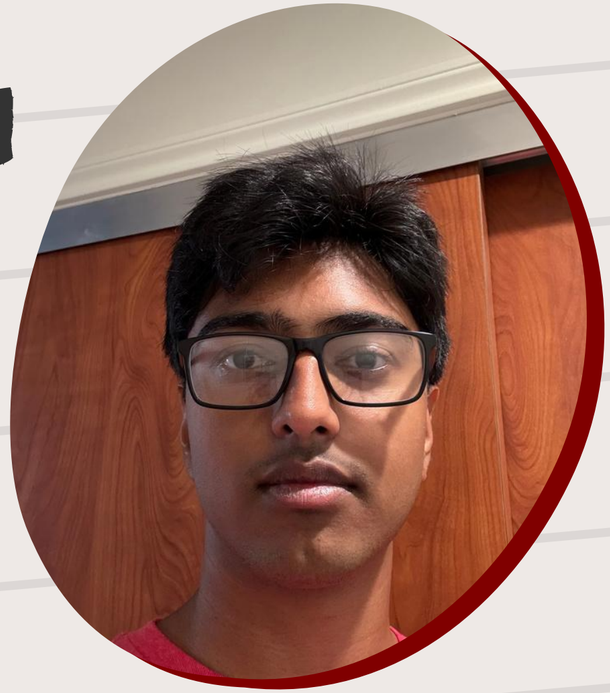
Welcome To QuantA&M!



Our Team



President

ADITYA



Treasurer

ISHAAN



Secretary

JOHNATHAN

Our Goals

General Understanding of Quantum Information

Professional Development

Proficiency with Quantum Tools

Org Expansion

Tracks

Engineering

Controls Signals

Hardware & Chip Packaging

Python, LTspice, KiCad

Physics

How Qubits Work

Noise & Errors

Python, QuTIP

CS/Math

Quantum Algorithms

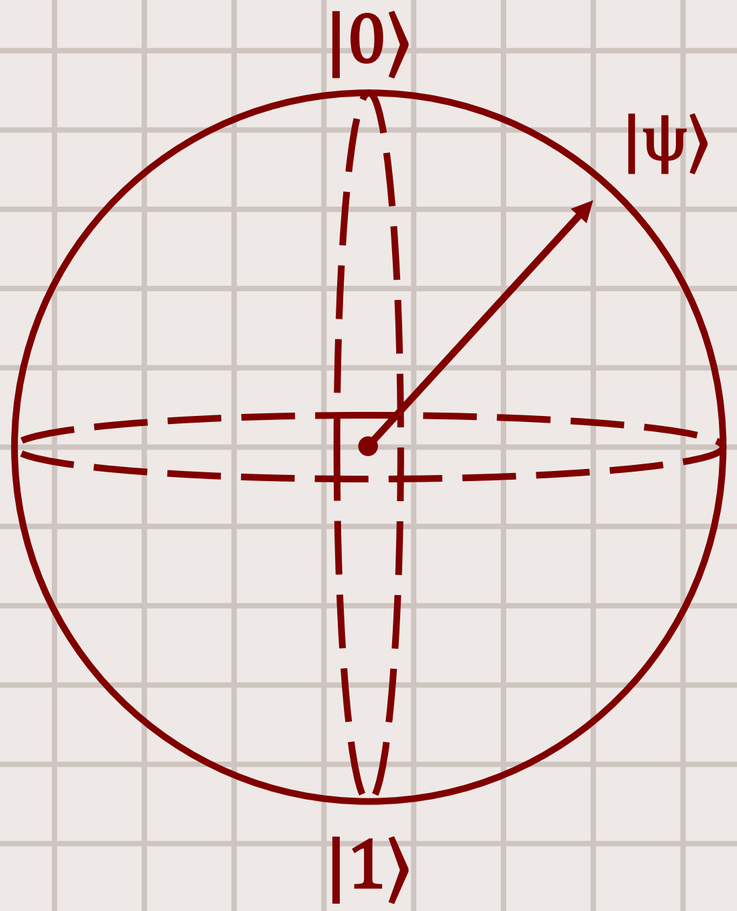
Quantum Logic & Games

IBM Qiskit, Google Cirq

Snapshot of Quantum Computing

Bit vs. Qubit

- **Classical bit:** always 0 or 1
- **Qubit:** a superposition of $|0\rangle$ and $|1\rangle$
- Measuring it gives 0 or 1, and the superposition collapses
- Every point on the Bloch sphere is a valid qubit state
- n qubits are described by 2^n amplitudes at once



Reading a Qubit

1. **Linear combination**
   $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
2. **Amplitudes**
   α and β are complex numbers
3. **Probability**
   $P(0) = |\alpha|^2,\ P(1) = |\beta|^2$
   $|\alpha|^2 + |\beta|^2 = 1$

Feedbac
k
Form



Thank You!



